

Complications after transfusion

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HAEMATOLOGY AND HAEMATOLOGICAL ONCOLOGY

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Lecture Outline

- ▶ Different kinds of blood products
- ▶ Acute adverse reactions after transfusion
- ▶ Chronic complications from transfusion
- ▶ How to prevent complications from transfusion

Blood products save millions of lives....



The first ever blood transfusion

....the last recipient was a man Mauroy with an “inveterate phrensy, occasioned by a disgrace he had received in some Amours”. Professor Denis hoped that “ the calf’s blood by its mildness and freshness might possibly allay the heat and ebullition of his blood.” The first two transfusions given to him apparently relieved his mania, although following the second his arm became hot, his pulse rose, sweat burst out over his forehead, he complained of pain in his kidneys, became sick to the stomach and passed black urine the next day....Nevertheless, the patient died the same night. The cause of death was examined. And it was concluded that the patient’s wife had been putting arsenic in his broth.

Reality is....



Blood products

- ▶ Red cells
- ▶ Platelet concentrates
- ▶ Fresh Frozen Plasma
- ▶ Cryoprecipitate
- ▶ Cryoprecipitate-reduced plasma
- ▶ Leucocytes (buffy coat)

Red cells

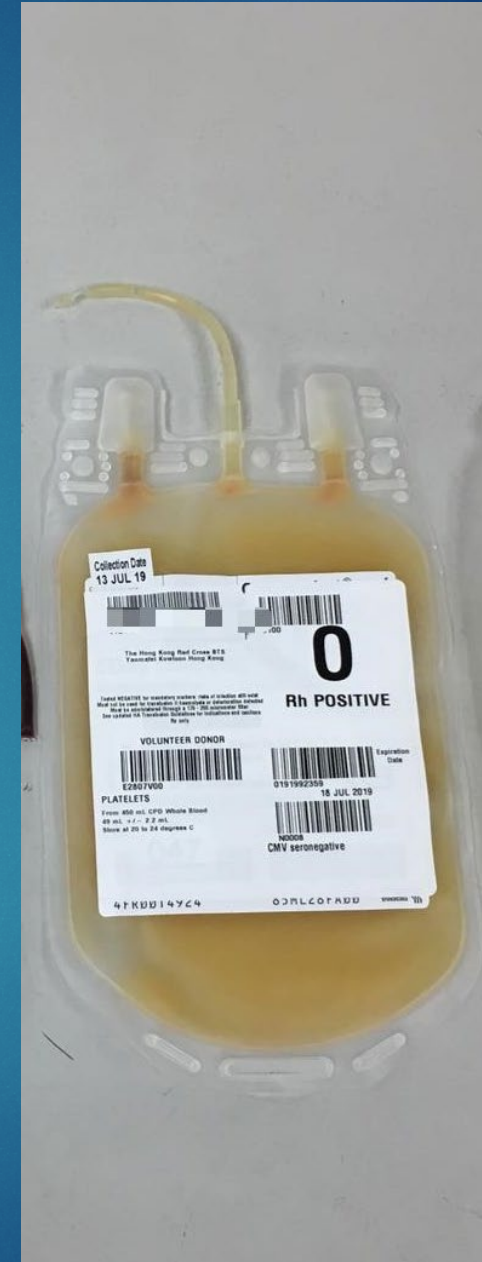
- ▶ Packed red cells with additive solutions (citrate-phosphate-dextrose-adenine)
- ▶ Stored at temperature 4C
- ▶ Shelf life 42 days
- ▶ Each red cell unit ~330ml



Source: Hong Kong Red Cross Blood Transfusion Service

Platelet concentrates

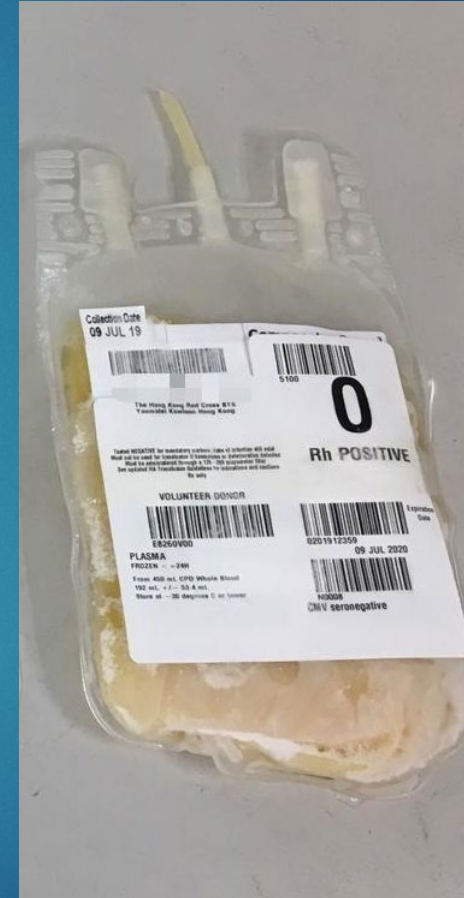
- ▶ Stored at room temperature (20-24C)
- ▶ Shelf life 4-5 days
- ▶ Requires agitation



Source: Hong Kong Red Cross Blood Transfusion Service

Fresh Frozen Plasma

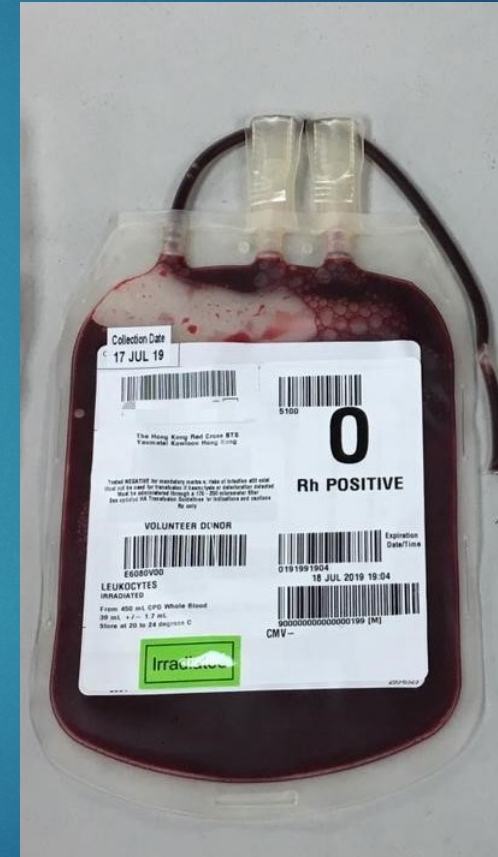
- ▶ Acellular component of blood
- ▶ Contains all the coagulation factors
- ▶ Stored at temperature -30C
- ▶ Must be thawed before use
- ▶ Each unit of FFP ~250ml



Source: Hong Kong Red Cross Blood Transfusion Service

Leucocytes

- ▶ Must be irradiated
- ▶ Stored at room temperature 20-24C
- ▶ Shelf life 8 hours



Source: Hong Kong Red Cross Blood Transfusion Service

What are the risks?

Volume

Infections

Immunological

Electrolytes

Vascular

Iron load

A common scenario in ward



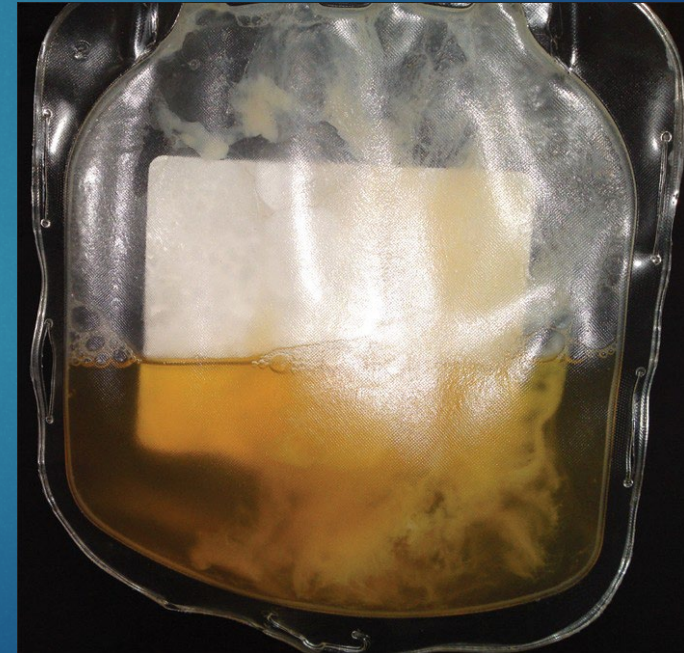
- Patient receiving red cell transfusion
- Develop kick of fever
- Now running a temperature of 38.5C
- Still have half unit of red cell not transfused
- Doctor, what should you do?

Fever during or after transfusion

- ▶ 1. Infection – bacterial contamination of the blood products
- ▶ 2. Hemolytic reaction – ABO incompatibility
- ▶ 3. Febrile non hemolytic transfusion reaction

Bacterial contamination of blood products

- ▶ Most common in platelet transfusion
- ▶ Inspect the blood product for any visible damage or discolouration
- ▶ Microbiological workup of patient
- ▶ Send remaining blood product for culture



ABO incompatible hemolytic transfusion reaction

- ▶ **Naturally occurring** IgM and IgG Anti-A and Anti-B
- ▶ Complement fixing antibodies
- ▶ Abrupt onset of “sense of impending doom”, flushing, fever, rigors, loin pain, vomiting, shock
- ▶ Haemoglobinuria
- ▶ May develop DIC and acute renal failure

ABO incompatible hemolytic transfusion reaction

- ▶ Activation of complements leading to intravascular hemolysis
- ▶ Free haemoglobin activates platelets, induce vascular inflammation, and binds nitric oxide causing smooth muscle dystonia
- ▶ Complement activation liberates potent anaphylatoxins, and degranulates mast cells
- ▶ Cytokines released from various phagocytes (TNF, IL-6, IL-8 etc)
- ▶ Activates coagulation system
- ▶ Renal failure maybe resulted from vasoconstriction, hypotension or DIC (NOT the free haemoglobin itself)

ABO incompatible hemolytic transfusion reaction

- ▶ ALWAYS A MEDICAL ERROR
- ▶ Can be fatal
- ▶ ALWAYS do Type and Screen properly
- ▶ ALWAYS give group-specific blood

ABO incompatible hemolytic transfusion reaction

- ▶ STOP the transfusion
- ▶ Intensive care often indicated
- ▶ Aggressive fluid resuscitation
- ▶ Alkaline diuresis
- ▶ Dopamine infusion to preserve renal perfusion



Febrile non-hemolytic transfusion reactions



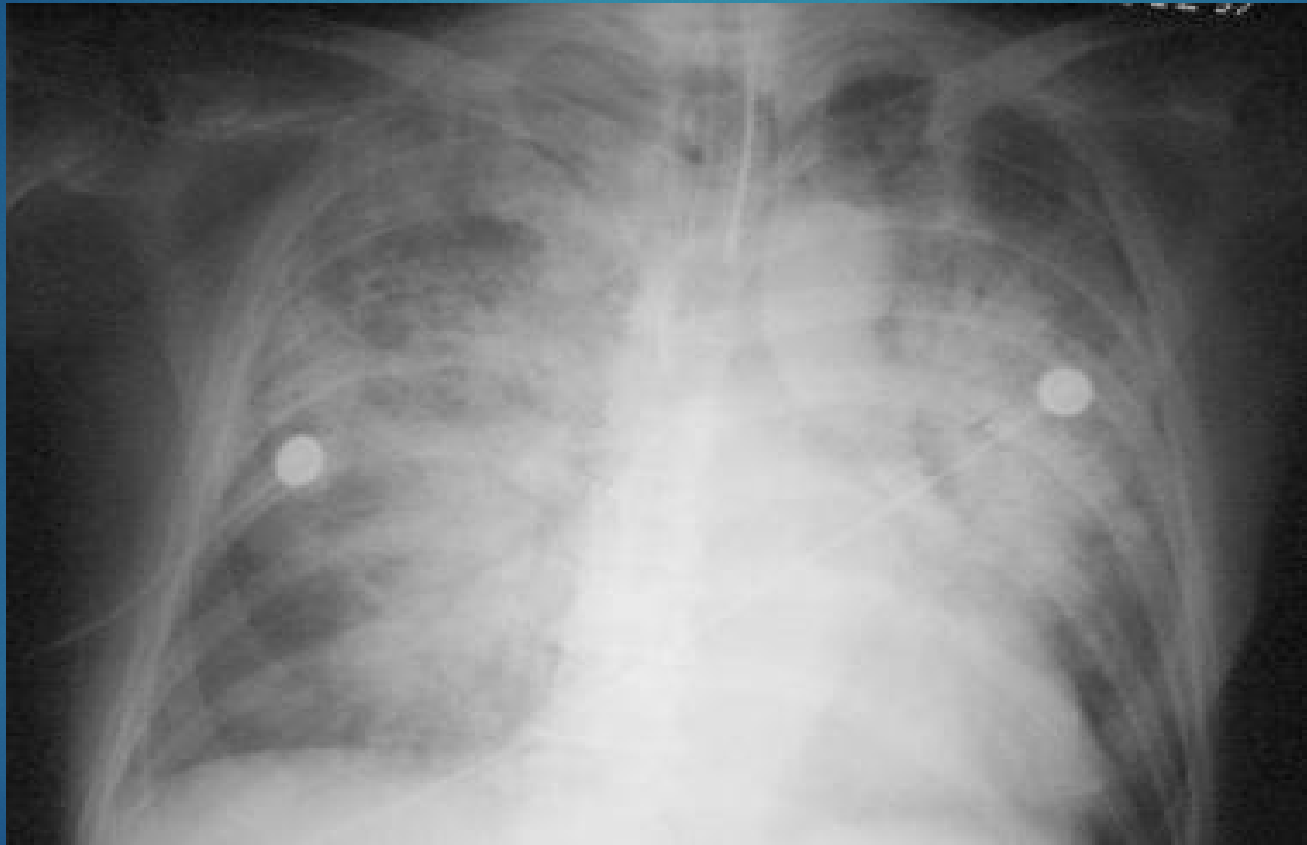
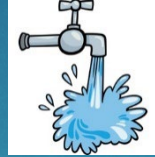
- ▶ Incidence: Red cells – 1-12.4%; platelets – up to 30%
- ▶ Rise of body temperature by 1°C or more during or within several hours of transfusion
- ▶ Array of signs and symptoms including chills, fever, dyspnea, hypotension, hypertension, rigors
- ▶ Leucoagglutinins bind to transfused leucocytes, activate monocytes and release cytokines with pyrogenic properties
- ▶ Leucoagglutinins could be anti-HLA class I or granulocyte specific antibodies

Febrile non-hemolytic transfusion reactions

- ▶ Febrile reactions could be caused by plasma factors other than leucoagglutinins in the recipients
- ▶ Cytokines including TNF α , IL-1 α , β , IL-6
- ▶ Correlates with number of residual leucocytes, storage time, storage temperature as well as rate of infusion of blood products
- ▶ Highest in platelet concentrates

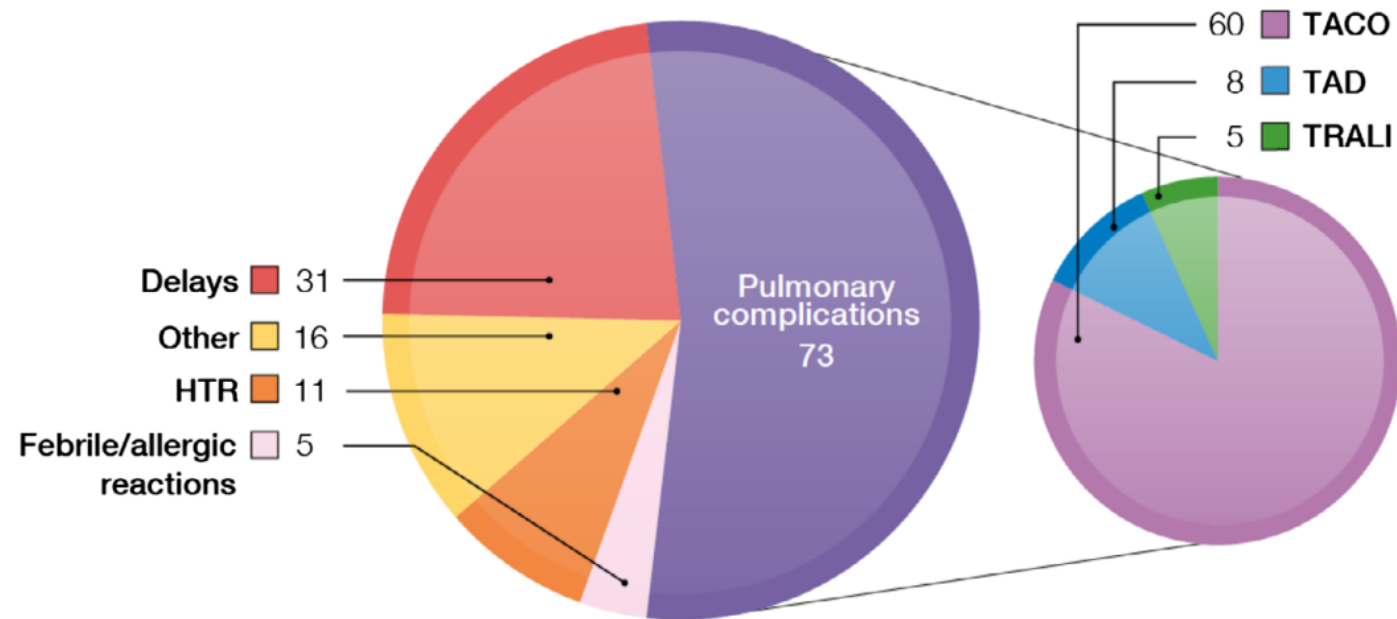
OTHER COMPLICATIONS FROM TRANSFUSIONS

Transfusion associated circulatory overload TACO



Occurs in 1-8% of transfusion

TACO is the leading cause of transfusion related mortality



'Other' includes 1 each for transfusion-transmitted infection, post-transfusion purpura, transfusion-associated graft-versus-host disease and anti-D related; there were 5 in the avoidable, over or undertransfusion category and 7 deaths related to other unclassified reactions

Fig. 3. Transfusion-related deaths 2010–2017, $n = 136$. (HTR = haemolytic transfusion reactions).

Transfusion associated circulatory overload

- ▶ Avoid unnecessary transfusion
- ▶ Transfuse ONE unit of red cell and reassess
- ▶ Transfusion rate
- ▶ Be careful in
 - ▶ Elderly
 - ▶ Renal impairment
 - ▶ Heart failure
- ▶ Judicious use of diuretics
- ▶ Close monitoring of hemodynamic and fluid status of patients during transfusion

Transfusion associated circulatory overload

- ▶ Plasma volume in adult 40ml/kg.
- ▶ To achieve haemostasis, 25-30% of factor level is adequate.
Therefore, replacing 1/3 to 1/4 of plasma volume of patient is enough
- ▶ Typical dose of FFP 10-15ml per kg
- ▶ Equivalent to 2 units of FFP in an average sized patient
- ▶ Significant volume challenge to patient
- ▶ Infuse 1-3ml/kg/hour with close monitoring of fluid status and judicious use of diuretics



Allergic reactions



- ▶ Incidence of mild allergic reactions 3%; anaphylaxis 1 in 20000 transfusion
- ▶ Ranged from urticarial to anaphylaxis
- ▶ Allergic reactions towards transfused proteins
- ▶ IgE mediated
- ▶ Beware of IgA deficient and haptoglobin deficient patients



Hypotensive reactions

- ▶ Hypotension, vasodilation due to reactions towards bradykinins, or prekallikrein activator in plasma products
- ▶ Bedside filtration of blood products exposes the plasma to charged synthetic surfaces
- ▶ Mainly a concern in patients on ACEI

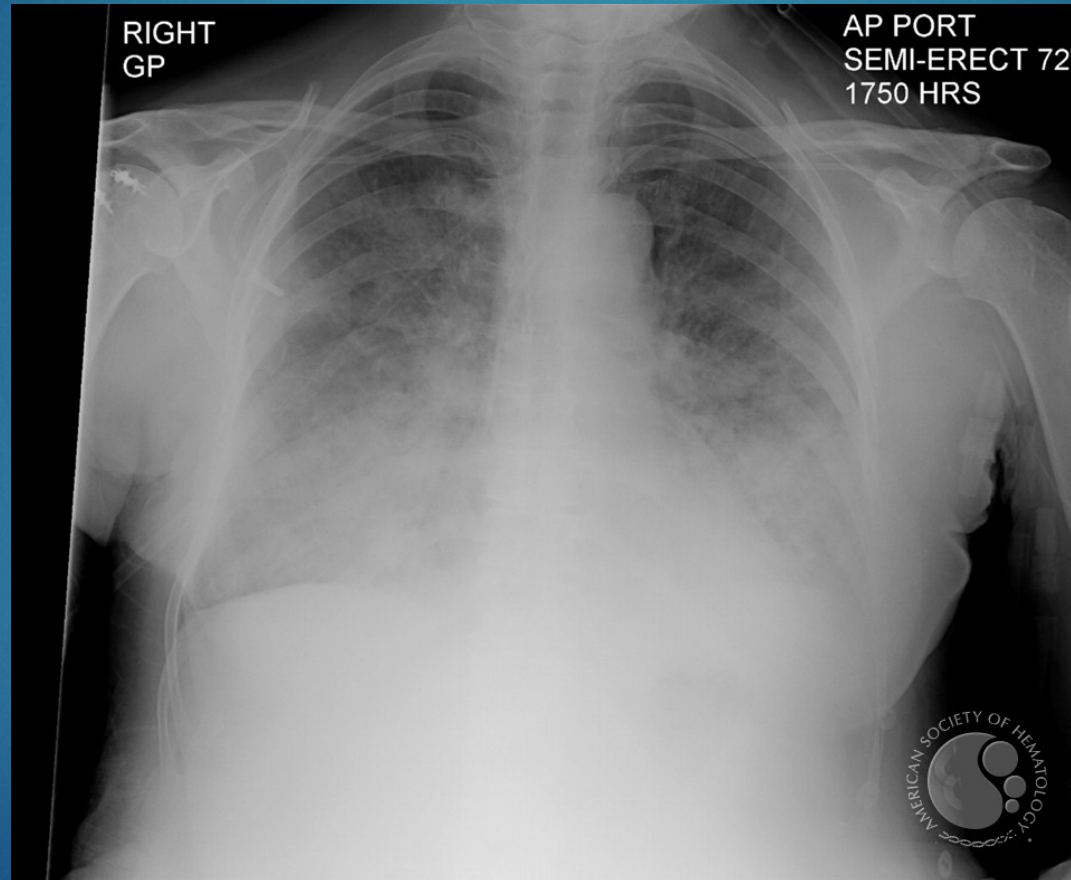
Delayed hemolytic transfusion reaction

- ▶ Delayed hemolytic transfusion reactions are usually due to minor blood group incompatibility e.g. Kidd antigen
 - ▶ Initial antibody titre too low to be detected in the T&S
 - ▶ Transfusion of incompatible red cells provokes anamnestic immune response
 - ▶ Onset of hemolysis 4-5 days after transfusion
 - ▶ Extravascular hemolysis
 - ▶ Fever, jaundice and rapid drop in Hb

Delayed hemolytic transfusion reaction

- ▶ Investigation
 - ▶ CBP, reticulocyte, manual film
 - ▶ Haptoglobin, methaemalbumin, direct/indirect bilirubin, LDH
 - ▶ Direct antiglobulin test, Indirect antiglobulin test
 - ▶ Extended red cell phenotype (of pre-transfusion sample)

Transfusion associated acute lung injury TRALI



TRALI

- ▶ Transfusion related acute lung injury
- ▶ Incidence 1 in 65 000 FFP infusions
- ▶ Non cardiogenic pulmonary edema
- ▶ Chills, fever, dyspnea and hypotension within one to two hours of transfusion. Onset of respiratory distress defined to be within 6 hours of initiation of transfusion.
- ▶ Radiological evidence of new bilateral pulmonary infiltrates in the absence of circulatory overload
- ▶ Pathogenesis believed to be due to leucocyte antibodies in donor plasma

Transfusion associated graft versus host disease TA GVHD

- ▶ Transfusion associated graft versus host disease
- ▶ Transfused allogeneic lymphocytes attack the recipient, causing marrow aplasia, diarrhea, rash and hepatitis
- ▶ 4 to 30 days after transfusion
- ▶ Mortality 90%
- ▶ Immunocompromised recipients e.g. preterm infants, bone marrow transplant recipients or patients with congenital immunodeficiency are at risk
- ▶ Immunocompetent recipients receiving blood products from donor who is homozygous at the HLA loci, of which the recipient shares the haplotype

Indication of irradiated blood products

Irradiated cellular blood components	Same as non-irradiated counterparts	For prevention of transfusion-related graft versus host disease in circumstances including: 1. Foetuses requiring intrauterine transfusion. 2. Neonates who have previously received intrauterine transfusion undergoing top-up transfusion till 6 months of age. 3. Patients with severe congenital cellular immunodeficiency. 4. Allogeneic haemopoietic stem cell transplantation patients. 5. Autologous haematopoietic stem cell transplantation/chimeric antigen receptor (CAR) T-cell therapy, from at least 7 days prior to bone marrow/stem cell/lymphocyte harvest until 3 months' post-transplantation/therapy. 6. Patients treated with purine analog drugs (e.g., fludarabine, cladribine, clofarabine, bendamustine) until 12 months after completion of therapy. 7. Patients treated with anti-T-cell treatment (e.g., anti-thymocyte globulin [ATG], alemtuzumab and other T-cell monoclonals) for aplastic anaemia or leukaemia from the initiation of treatment till 12 months after completion of therapy. 8. Patients receiving transfusion from close relatives. If there is an urgent need for transfusion, this should not be withheld if no irradiated unit is available. Leukodepleted (performed pre-storage or by bedside inline filter) blood products and older (stored > 2 weeks) red cells/whole blood units should be used.
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Following massive transfusion



Hypothermia

- ▶ Transfusion of large volume of blood could lower the core body temperature to as low as 27.5C
- ▶ Fatal complications from hypothermia reported
- ▶ Cause arrhythmia, cardiac arrest, platelet dysfunction and coagulopathy
- ▶ Massive transfusion must be administered via suitable blood warmer

Citrate toxicity

- ▶ Citrate is present in stored red cell as anticoagulant
- ▶ Massive rapid transfusion of red cells could lead to citrate toxicity
- ▶ Risk is higher in infants and patients with liver impairment
- ▶ Citrate chelates calcium, lowers ionized calcium level, leading to QT prolongation and arrhythmia
- ▶ Essential to replace calcium with 10% calcium gluconate (10ml Ca gluconate for every 1 litre of citrated blood is transfused)



Potassium load



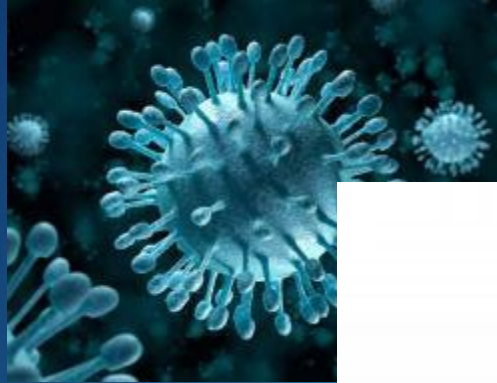
- ▶ Potassium is released from red cells with storage
- ▶ Extracellular potassium in a unit of red cell stored for 28 days = 10.2mmol
- ▶ Risk of hyperkalaemia in infants and patients with renal impairment



Long term complications

Infectious Risk of Transfusion

Hepatitis C



Hepatitis B



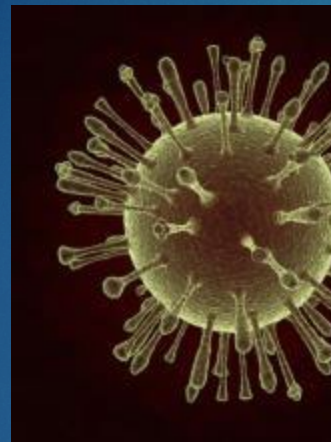
HIV



Malaria



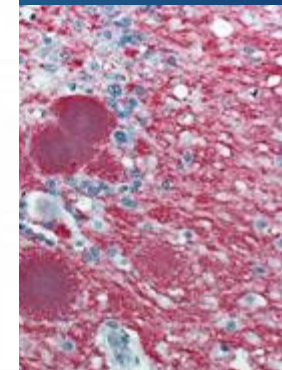
Zika



HTLV



Cellar slide
vCJD



Nile Virus



Prevention of transfusion transmitted CMV infection – use of CMV negative product

Cytomegalovirus (CMV) negative cellular blood components	Dosage according to individual cellular blood components	1. Elective transfusions during pregnancy (not during labour, delivery, miscarriages or termination of pregnancy).[#] 2. CMV-antibody-negative recipients of allogeneic stem cell grafts.* 3. Intrauterine transfusions. 4. Neonates with birth weight < 1.5kg [#]In emergency situations, if CMV negative cellular blood components are not available, life-saving transfusion should not be withheld and leukodepleted blood components can be used to avoid delays. *If CMV negative cellular blood components are not available, leukodepleted blood components can be used.
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Haemosiderosis

1 unit of blood = 200mg iron

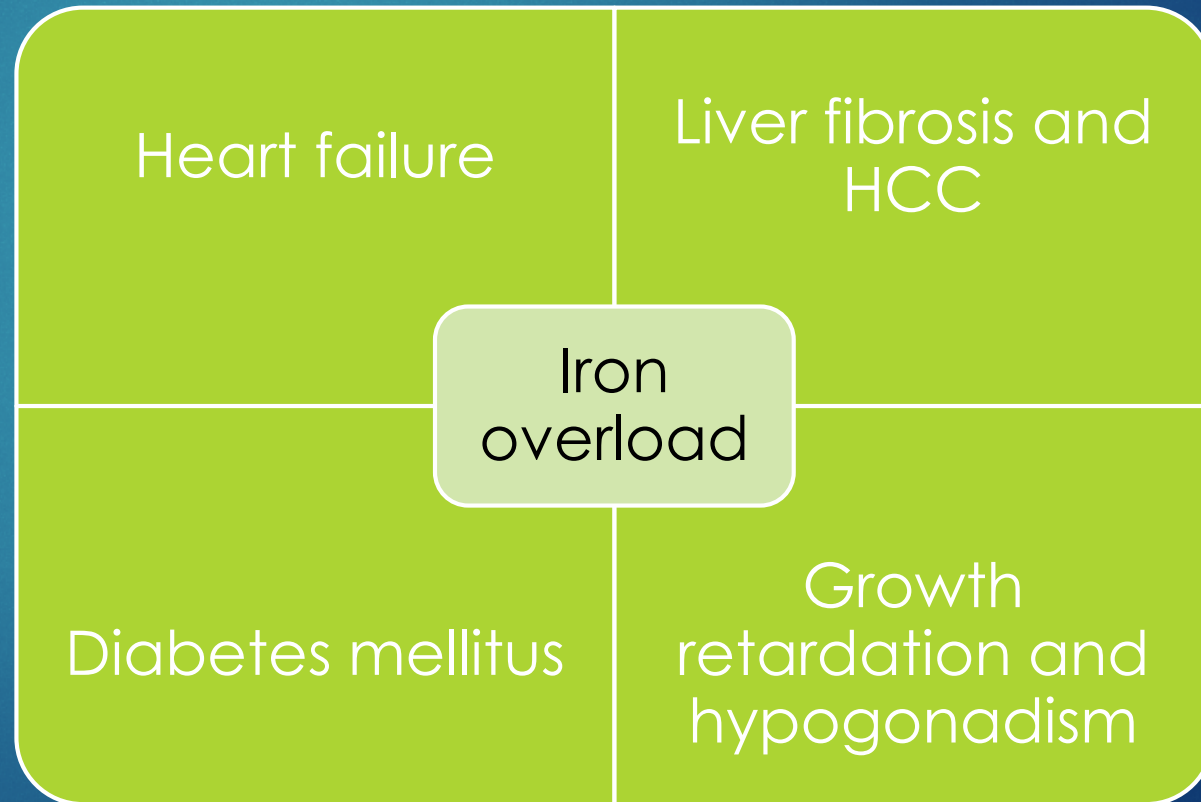
Daily excretion of iron 1mg



ET

Transfusion haemosiderosis

- ▶ Iron accumulation in liver, endocrine organs and heart

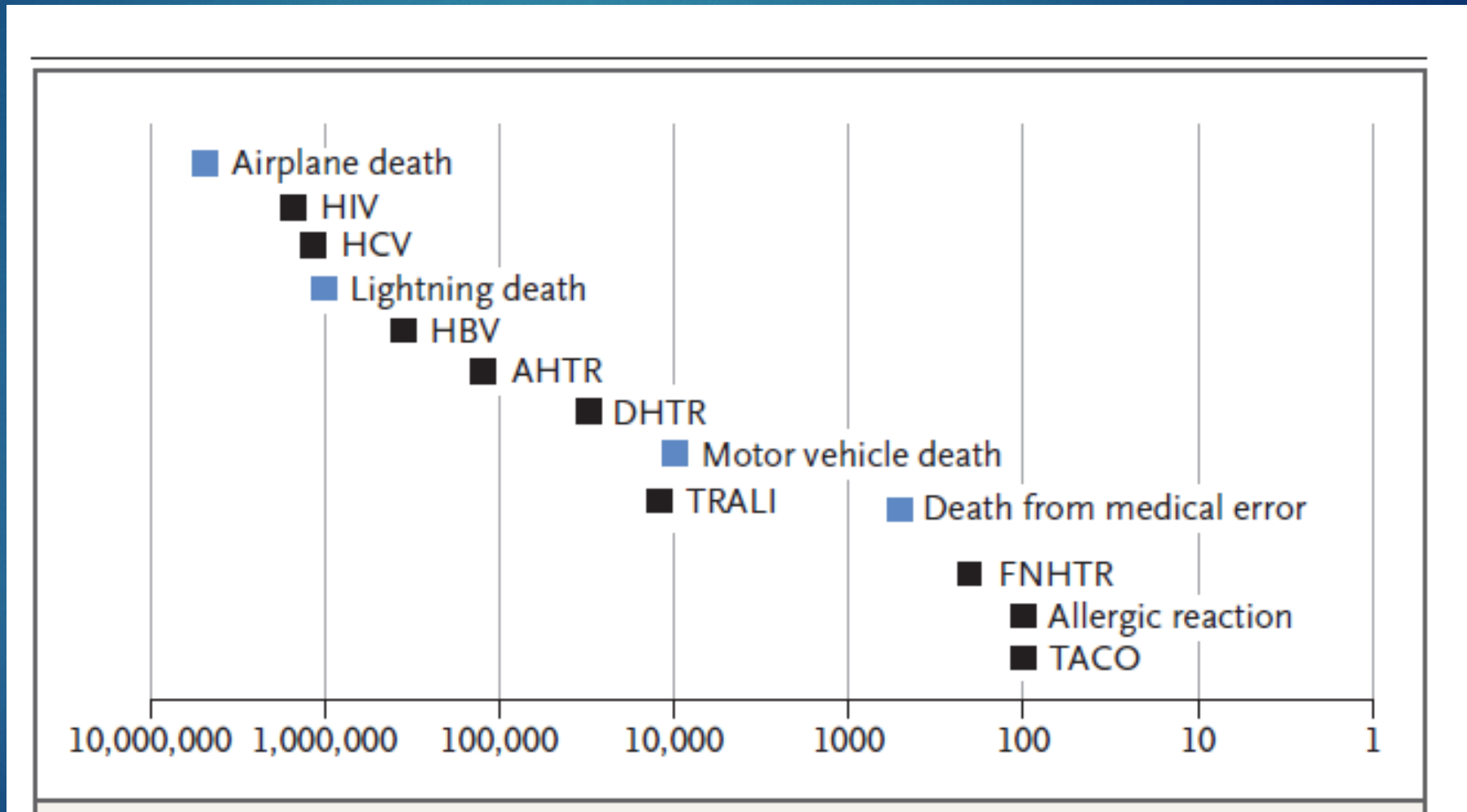


Transfusion haemosiderosis

- ▶ Ferritin
- ▶ MRI T2* liver and heart
- ▶ Liver biopsy

- ▶ Treatment by iron chelation

Risk of various adverse effects of red cell transfusions



Reality is....



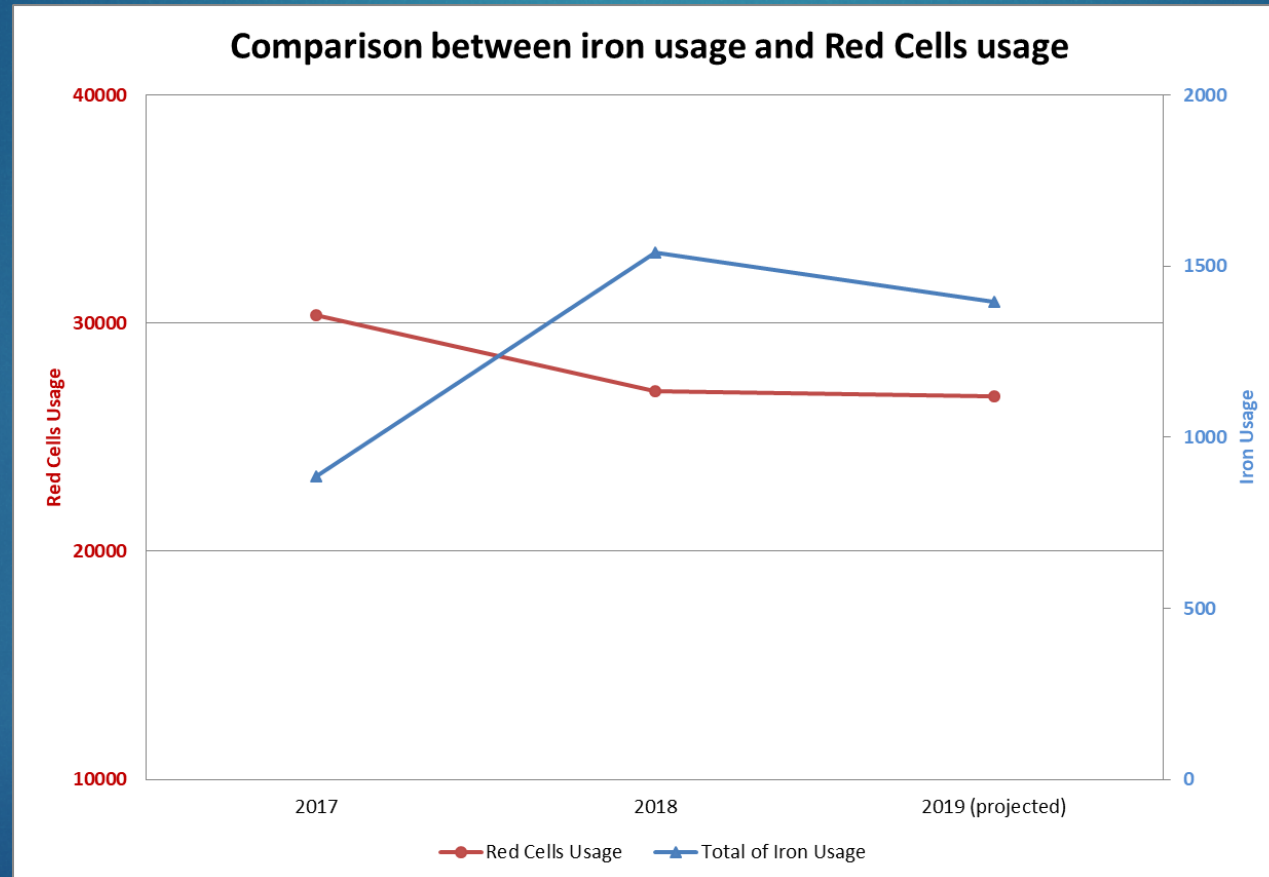
How to prevent it

- ▶ Voluntary donation, donor deferral to reduce risk of infection transmission
- ▶ ? Universal pathogen inactivation
- ▶ Prestorage leucocyte reduction (aim less than 0.5×10^9 leucocytes per unit)
- ▶ Bedside filtration
- ▶ Use of premedication e.g. antihistamine and steroid
- ▶ Use only male donors in preparation of FFP
- ▶ Irradiation of blood products before transfusion in all immunocompromised patients

How to prevent it

- ▶ The safest transfusion is not to transfuse at all
- ▶ Alternatives to allogeneic blood
 - ▶ Iron
 - ▶ Acellular volume expanders

How to prevent it



Courtesy Dr. Rock Leung

Patient Blood Management

1st Pillar

OPTIMIZE
HEMATOPOIESIS

2nd Pillar

MINIMIZE
BLOOD LOSS

3rd Pillar

OPTIMIZE
PATIENT-SPECIFIC
TOLERANCE OF
ANAEMIA



SAVE BLOOD, SAVE LIVES

Transfusions are one of the most overused treatments in modern medicine, at a cost of billions of dollars. Researchers are working out how to cut back.

BY EMILY ANTHES

Thank you